

Sina Ronaghi

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LinkedIn | ResearchGate | GitHub | Online Portfolio

PROFILE SUMMARY

Ph.D. Electrical Engineer bridging **metrology**, **embedded systems**, and **data science**. Experienced in **project and operations management** in international R&D consortia (Erasmus+, HORIZON) and executing full-lifecycle product development from startup ideation to academic research. Highly skilled in applying machine learning and robust statistical frameworks (JCGM/GUM) to optimize energy and measurement infrastructures.

CORE COMPETENCIES

Hardware & Metrology: A/D Circuit Design (Altium, LTSpice), Embedded Systems (ARM Cortex M/Arduino), Instrumentation (DAQ, DMMs, Oscilloscopes, Spectrum/Logic Analyzers), JCGM (GUM), Characterization & Calibration, ISO 9001, ISO/IEC 17025, IEC 61000, IEC 62052/53/56, EN 50160.

Software & Data Science: Python (scikit-learn, PyVISA, Pandas), MATLAB, NI LabVIEW/TestStand, C/C++, R, Node.js, SQL/No-SQL, Supervised/Unsupervised Learning, Time Series Forecasting, PCA, MC Simulation.

Frameworks & Leadership: Agile (Jira, Trello), LEAN & Kaizen, R&D Project Management, Cross-Functional Team Leadership, Git/GitLab, L^AT_EX, Public Speaking.

PROFESSIONAL EXPERIENCE

Project Manager (EU) / Post-Doc Researcher

Nov 2025 – Present

Politecnico di Milano

Milan, Italy

- Led the adoption of advanced VT/CT characterization methods under non-linear conditions, applying advanced numerical techniques in MATLAB, NI LabVIEW, and Python to cut required test repetitions by 70% under EN 50160 MV/HV levels.
- Served as the accountable lead for quality and compliance, enforcing ISO/IEC 17025 and ISO 9001 across HV/MV precision test and calibration labs.
- Directed the EU Erasmus+ PEMS-ML program, coordinating 3 partner institutions and 15+ contributors and ensuring milestone delivery through Agile and LEAN practices.
- Coordinated internal contributions to HORIZON 6 proposal drafting, aligning work across 18+ EMEA partners, including Imec, Infineon, and Siemens.

R&D Associate Manager - EV Sensor Fusion Project

Nov 2024 – Oct 2025

Politecnico di Milano

Milan, Italy

Recruited to lead a research project on electric vehicles (EV), while concurrently finalizing my Ph.D. dissertation.

- Led the research strategy for EV attitude-estimation, guiding the design of an ARM-based embedded testbench in C and new sensor-fusion methods, achieving $\sim \pm 1.5^\circ$ angular-error performance.
- Directed the shift to standardized, automated test workflows, implementing NI LabVIEW/TestStand pipelines that cut testing time by 50% and improved repeatability.
- Owned statistical validation for a 3-month EV test-drive campaign, performing time series anomaly detection in Python that ensured sensor integrity and revealed key efficiency/behavior correlations.

Ph.D. Researcher - Energy Metering Anomaly Detection

Nov 2021 – May 2025

Politecnico di Milano

Milan, Italy

- Performed supervised and unsupervised data-driven analyses (in Python) on Italy's Smart Metering infrastructure to support legal metrology requirements in accordance with IEC 62052 and IEC 62053.
- Developed a robust and traceable telemetry pipeline for 2G Smart Energy Meters over PLC Band C (IEC 62056, CEI TS 13:82-84), enabling 3+ years of continuous data collection from five nodes with $< 5\%$ data loss using IIoT (M2M) architectures deployed on AWS.
- Researched, refined, and published advancements to metrological guidance documents (JCGM), with a focus on conformity-assessment and decision-rule frameworks such as GUM 106.

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| Technical Consultant
<i>Panak IoT Solutions LLC. [startup]</i> | Jun 2019 – Aug 2019
<i>Ahvaz, Iran</i> |
| <ul style="list-style-type: none"> • Developed cloud-integrated (IIoT) test setups for precision agriculture machinery that achieved < 1% packet loss during extended (> 3h) real-time field tests via ESP32 embedded hardware, Node-RED, and SMQTT. • Developed NI LabVIEW interfaces for real-time monitoring and closed-loop control of industrial synchronous motors, implementing PID actuation controls. | |
| Backend Developer (RESTful APIs)
<i>Diacio Game Studios LLC. [startup]</i> | Sep 2018 – Aug 2019
<i>Ahvaz, Iran</i> |
| <ul style="list-style-type: none"> • Ensured lossless packet delivery via robust RESTful API structures in multiplayer online mobile games for user authentication, in-app payments, and leaderboards, utilizing Node.js and MongoDB. • Engaged in strategic alignment and business direction consultancy with core stakeholders for long-term planning that resulted in 18% growth in the company's fund acquisition YoY through start-up pitches. | |
| Co-Founder & CEO
<i>Atiyeh Hoshyar LLC. [startup]</i> | Dec 2017 – Aug 2018
<i>Ahvaz, Iran</i> |
| <ul style="list-style-type: none"> • Co-founded an IoT startup and secured 60% of total funding through targeted pitches and incubator programs, accelerating early product development and market validation. • Scaled engineering capacity by 40% in the first year by recruiting, onboarding, and leading a 5-person R&D team specializing in electronic design and embedded hardware/firmware. • Filed a patent in Home Automation and Remote Monitoring, integrating secure M2M communication (SMQTT) into ARM-based embedded hardware/firmware architectures. | |

EDUCATION

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| Politecnico di Milano
<i>Ph.D. in Electrical Engineering (Summa Cum Laude)</i>
<i>Thesis: Data-driven approaches for anomaly detection in energy metering</i> | Nov 2021 – May 2025
<i>Milan, Italy</i> |
| Politecnico di Milano
<i>M.Sc. in Electrical Engineering (106/110)</i>
<i>Thesis: Real-time motion capturing using inertial measurement units and applied embedded systems development</i> | Sep 2019 – Oct 2021
<i>Milan, Italy</i> |
| Shahid Chamran University of Ahvaz
<i>B.Sc. in Electrical Engineering (14.56/20)</i>
<i>Thesis: Embedded Systems Design for Test Automation in AgriTech</i> | Sep 2013 – Jul 2018
<i>Ahvaz, Iran</i> |

HONORS & MEMBERSHIPS

- **Awards:** IEEE AMPS 2023 - Best Presentation Award; Open Subject Scholarship for PhD Studies.
- **Speaking:** Invited speaker at CNI Milano Digital Week 2024 for a 3-year regional energy monitoring project.
- **Committees:** Technical Program Committee (TPC), IEEE MetroXRAINE (2024–2026).
- **Memberships:** GMEE (Gruppo Misure Elettriche ed Elettroniche), IEEE Instrumentation & Measurement Society, IEEE Young Professionals (Member since 2021).

CERTIFICATIONS & LANGUAGES

- **Certifications:** LEAN Management Fundamentals, Probability and Data with R, Python Data Structures, Data Scientist's ToolBox (Coursera); NI LabVIEW Object-Oriented Programming (Emerson); Signal and Power Integrity for High-Speed PCBs (Polimi); EU Driving License (B).
- **Languages:** English (Professional), Italian (Working Proficiency).